



DATASHEET

Quantum 6³⁸ / 6^{38T}

Compact without Compromise

OVERVIEW

The DiGiCo Quantum 6³⁸ / 6^{38T} is a dual engine, 256 channel mixing console with 38 physical faders and 3 x 17" LCD high-resolution touchscreens. Combining the extraordinary might of large-scale FPGAs, the Quantum 6³⁸ / 6^{38T} redefines digital mixing consoles.

KEY FEATURES

256 Input Channels with full processing

128 Aux/Sub-Group busses with full processing

48 x 48 Matrix with full processing

64 Insertable Mustard processing strips

128 Nodal processing instances

16 Spice Rack Slots

True Solo function

User programmable macros (including channel specific buttons)

Dual redundant engines

Assignable channel Layout

Redundant hot-swappable PSUs as standard

Snapshots for seamlessly changing many parameters at once

2 DMI Slots per engine to expand the I/O as desired

Offline software

iPad control

Theatre specific software option



DiGiCo Quantum

Quantum is developed with seventh-generation FPGA devices that further expand audio processing power and ultimately allows DiGiCo to provide its users with an unrivaled amount of additional flexibility.

Quantum . Power . Connectivity . Flexibility



TECHNICAL SPECIFICATIONS

WORKSURFACE

38 x 100mm touch-sensitive, motorised faders
3 x 17" LCD high-resolution touchscreens
3 x 6.8" Bank Switcher Screens
15 x 1.3" Full colour TFT LCD displays
38 x 1.3" Touch sensitive full colour TFT LCD displays
3 x 14" Custom mounted high-resolution LCD Meterbridge screens
2 x ¼" Headphone sockets
2 x 3.5mm Headphone sockets
2 x Headphone rests
1 x USB 2.0 port
1 x XLR Talkback input
1 x Microphone holder
Keyboard Light Bar
Integrated Light Bar

OPTIONS

Upgrade to Dual Loop Optocore (HMA, OpticalCon or ST)
Upgrade to SingleMode Optocore
Theatre Specific Software
Flightcase
Compatible DMI Cards: ADC / AES / AMM / AVB / DAC / Dante / Dante64@96 / Hydra 2 / KLANG / MADI B / MADI C / ME / Mic / Waves

REAR

2 x Redundant, hot-swappable PSUs
8 x XLR 32 bit Mic/Line Inputs
8 x XLR 32 bit Line Outputs
4 x XLR AES/EBU Inputs (8 x channels)
4 x XLR AES/EBU Outputs (8 x channels)
1 x GPI DSub37 (16 inputs)
1 x GPO DSub37 (16 outputs)
1 x MIDI In/Thru/Out (5 pin DIN)
1 x SMPTE I/O (XLR) plus level control

Per Engine:

1 x MultiMode Optocore Interface (expandable to 2)
On Board Dante Sockets (Primary & Secondary)
1 x Fourier Interface Card as standard
2 x DMI Slots (up to 64 I/O per slot)
4 x 1Gb Ethernet ports (switched together)
8 x MADI BNC I/O @ 48kHz, 4 interfaces at 96kHz
1 x UB MADI (USB Type B Audio I/O interface for recording and playback of up to 48 channels)
2 x USB 2.0 ports
2 x USB 3.0 port
1 x Word Clock I/O BNC (2 BNCs)
1 x AES/EBU XLR Sync I/O (2 XLRs)
1 x DisplayPort Output

TECHNICAL SPECIFICATIONS

SIGNAL PROCESSING

256 Input Channels (Mono)

- Main & Alternative input
- Analogue Gain
- Phase Inversion Control
- Gain Tracking
- Digital Trim (-40dB to +40dB)
- Variable Delay (0ms to 1.3s)
- DiGiTube
- HPF/LPF (-24dB/Oct)
- 4 Band Parametric EQ / Dynamic EQ
- DYN 1: Compressor, Multiband Compressor, Desser
- DYN 2: Gate, Duck, External Input Compressor
- EQ/Dyn Order Control
- 2 Insert Points per Channel
- Channel Mute & Hard Mute
- Channel Direct Outputs

128 Aux/Sub-Group Busses

- Phase Inversion Control
- Digital Trim (-20dB to +60dB)
- Variable Delay (0ms to 1.3s)
- DiGiTube
- Merge Input
- Tone Generator
- HPF/LPF (-24dB/Oct)
- 8 Band EQ: 4 Band Parametric EQ and 4 Band Parametric or Dynamic EQ
- DYN 1: Compressor, Multiband Compressor, Desser
- DYN 2: Gate, Duck, External Input Compressor
- EQ/Dyn Order Control
- 2 Insert Points per Channel
- Channel Mute & Hard Mute

1 LR/LCR/LCRS/5.1 Master Buss (with full processing)

48 input x 48 output Full Processing Matrix

36 Control Groups (CGs)

2 Solo Busses

32 x 32-band Graphic EQ's

48 x Internal Stereo FX Processor

- Delays
- Audio Enhancer
- Choruses
- Pitch Shifters
- Reverbs

16 x Internal Spice Rack Slots

- 6 band Dynamic Multiband Compressor/Expander
- 6 band Dynamic EQ

64 x Mustard Processing Strips

- Tube Emulation
- HPF/LPF (-24dB/Oct)
- 4 Band EQ: 2 Band Parametric EQ and 2 Band Parametric EQ or All Pass Filters
- DYN 1: Classic RMS/Peak Compressor, Vintage VCA Compressor, Optical Compressor, FET Limiter, Levelling Amplifier
- DYN 2: Gate, Ducker, MSE

128 x Nodal Processing

- 4 Band Parametric EQ / Dynamic EQ
- DYN 1: Compressor, Multiband Compressor, Desser
- DYN 2: Gate, Duck, External Input Compressor

DiGiTuBes available on every channel and Buss

Dynamic EQs available on every channel, Buss and Nodal Processor

Multiband Compressors available on every channel, Buss and Nodal Processor

True Solo function

Virtual Soundcheck



In a world as competitive for engineers as it is for console owners, you want the best tools you can lay your hands on. You also want a console and audio tools as well thought out for every major application as they are designed for the art and science of sound engineering.



QUANTUM 638





A&E SPECIFICATION

The DiGiCo Quantum 6³⁸ / 6^{38T} shall have 38 faders split into 3 worksurface sections and two dedicated master faders. The left, centre and right worksurface sections shall have 6 layers of 6 banks. All faders can be assigned to control any channels of any type. The console shall be capable of 256 input channels, 128 Aux/Sub-group Busses, a LR/LCR/LCRS/5.1 Master Buss, 36 VCA style or mute group style Control Group channels, 2 Solo Busses, and a 48 input x 48 output full processing Matrix. All processing paths shall have full processing including Tube emulation, Dynamic EQ and Multiband Compression. Tube emulation, Dynamic EQ and Multiband Compression shall be available on every channel and Buss on the console. All channel processing shall be internal and FPGA-Based.

An internal FX rack shall allow users to pick from 34 different FX. Up to 48 stereo FX can be added, comprising of up to 32 floating point reverbs and up to 48 delay/chorus/pitch/enhancer effects. An internal set of 32, 32-band Graphic EQs shall also be accessible. There shall be an additional processing rack called the Spice Rack, allowing up to 16 mono effects. There shall also be 64 insertable Mustard Processing Strips. The position of these, and of inserts, shall be chosen from pre-processing, pre-EQ/dynamics, mid-EQ/dynamics, pre-fader or post-fader. 128 instances of nodal processing shall also be available. These shall allow EQ and dynamics (including Dynamic EQ and Multiband Compression) to be applied to the aux node of a channel.

Three 17" (42cm) high-resolution, high-brightness LCD touch screens shall be provided to show the channel strips. There shall be a physical button to switch the view of the centre screen to show the master view.

All three screens shall have a dedicated hardware channel strip, allowing for control over input parameters, filters, EQ, dynamics, insert points, aux sends, and two user assignable encoders and buttons. Each screen shall also have 2 rows of rotary encoders to control various channel parameters. The master section shall have physical controls to allow control over some snapshot functions, basic Solo functions, and a button to control which engine is being viewed. There shall also be 6 banks of 10 user-assignable LCD macro buttons on the worksurface. The user shall also be able to program macros that can be triggered with fader movements, GPI, MIDI and keyboard functions. Each fader shall also have two additional macro buttons ('a' and 'b'). The assignment of these shall change with channel assignment. The console shall have three meterbridge screens to show channel metering. The right meterbridge shall have 1 USB port.

The rear panel shall have 8 32-bit Mic/Line inputs, 8 32-bit line outputs, 4 AES/EBU inputs (8 channels) and 4 AES/EBU outputs (8 channels). It shall also have 1 DSub37 GPI and 1 DSub37 GPO (16 inputs and 16 outputs), 1 set of MIDI In, Thru and Out, and a set of XLR SMPTE LTC I/O. It shall also have 2 redundant and hot-swappable power supplies. Each engine shall have a MultiMode Optocore interface, providing 504 additional audio paths at 48kHz and 96kHz. The Optocore connection type shall be chosen from HMA, OpticalCon or ST. They shall also have a Fourier Interface Card, 2 DMI slots, 4 switched ethernet ports, 8 sets of MADI I/O, giving up to 8 MADI interfaces or up to 4 redundant MADI interfaces at 48kHz or 4 interfaces at 96kHz, 2 USB 2.0 and 2 USB 3.0 ports, 1 DisplayPort output, external Wordclock I/O and AES sync.

There shall be option to add a second Optocore loop to each engine. This shall give an additional 504 audio paths at 48kHz and 96kHz. The Optocore interface can also be upgraded to SingleMode. There shall be a Theatre Software option that shall provide Auto Update, Aliases, Players, Advanced CG programming tools and Matrix nodal delays.

The dimensions of the Quantum 6³⁸ / 6^{38T} shall be: 1555 (w) x 957 (d) x 535 (h) mm

The weight of the Quantum 6³⁸ / 6^{38T} shall be: 93kg

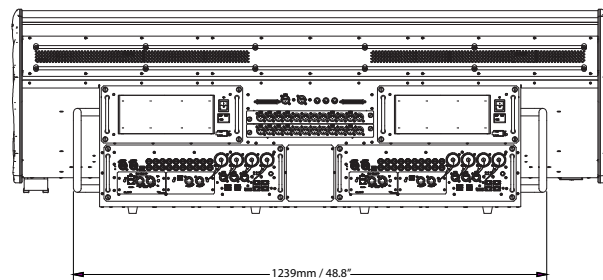
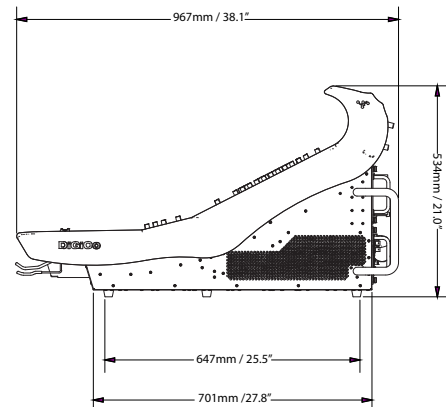
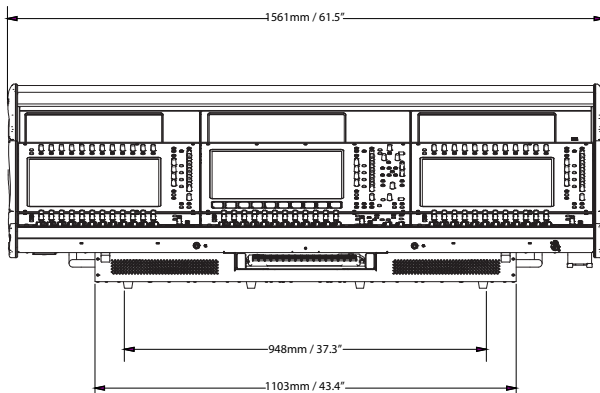
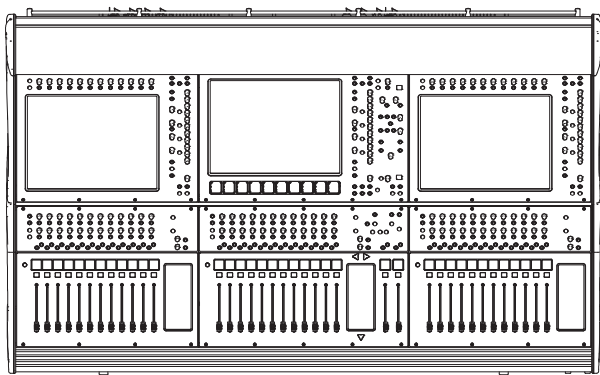
The DiGiCo Quantum 6³⁸ / 6^{38T} shall be supplied with a dust cover.

In a world as competitive for engineers as it is for console owners, you want the best tools you can lay your hands on. You also want a console and audio tools as well thought out for every major application as they are designed for the art and science of sound engineering.



LINE DRAWING

All dimensions in mm



PHYSICAL

Dimensions: 1555 mm (w) x 957mm (d) x 535mm (h)

Weight: 93kg (210kg with optional flightcase)

Flightcase: 1690mm (w) x 1312mm (d) x 636mm (h) (Optional)

Power Requirements: 100-240V, 50-60Hz 580VA

Redundancy: Internal removable engines x 2 internal hot-swappable PSUs x 2

Product Code: X-Q638-WS | X-Q638-WS-D or X-Q638-WS-F | X-Q638-WS-D-F

AUDIO SPECIFICATIONS

Frequency Response (Mic In to Line Out): +/- 0.2dB, 20Hz – 20kHz

THD+N (Mic In to AES Out): <0.001% @ 1kHz, 0dB gain, -10dBFS out, 22Hz-20kHz

THD+N (AES In to Line Out): <0.0005% @ 1kHz, +16dBu out, 22Hz-20kHz

Channel Separation/Crosstalk (Mic In to Line Out): >100dB @ 10kHz

Residual Output Noise (Line Out): <-98.5dBu A-weighted, 22Hz – 20kHz

Equivalent Input Noise/EIN (Mic In): <-129dBu A-weighted, 22Hz – 20kHz

Latency (48kHz): 0.86ms (Mic In to Line Out, no channel processing)

Latency (96kHz): 0.43ms (Mic In to Line Out, no channel processing)

Maximum Output Level: +22dBu

Maximum Input Level: +22dBu

DiGiCo HQ

No5 The Distillery, Silverglade Business Park, Leatherhead Road,
Chessington, Surrey, KT9 2QL, United Kingdom
info@digiconsoles.com

©DiGiCo 2026. All brand and product names are copyright to their respective owners
E&OE

DiGiCo
www.digico.biz